

Businesses can take advantage of Ethical Hackers who are generally referred to as White Hat Hackers to help them find and exploit the vulnerabilities and weaknesses in their systems. Ethical Hackers are information security experts who legitimately penetrate the system or the network on behalf of the company to find loopholes in the existing setup. Having ethical hackers on board can help organizations to anticipate what the hacker might do and take precautionary steps to reduce the risks.

This chapter will cover the different methods that can be used to combat the different attacks, and finally, a detailed list of auditing process for all the mobile phones and PCs to stay free from attacks.

Methods to combat different hack attacks

In this section, we will take a look at the techniques that can be used to combat different types of hacks explained in this book.

1. SQL Injection

While a successful attack on a SQL database can give great advantage to the hackers from being able to modify the contents of a website to retrieve sensitive information, it's important that business owners ensure the security of their data. One of the simplest ways is for them to try the attacks by using automated SQL injection tools on their web application and see if there are any vulnerabilities. Here are the precautionary safety steps that website owners can take to prevent SQL injection attacks.

- **Validate all user inputs** – Make sure all user submitted inputs are validated to ensure there are no special characters such as `'` are passed as arguments in the SQL query. For example, all the special characters in the email and name field should be properly sanitized and filtered, and the phone number field should only contain digits, and so on.
- **Use Parameterized Queries and Stored Procedures** – It's an advisable practice to use Stored Procedures, which automatically adds an extra layer of security on the database. The advantage of using the Stored Procedure is that the SQL code is first written and stored in the database server and only then called from the web application. Using the stored procedure will make the web application assume the user inputs as data to be operated on rather than SQL code that has to be executed.